

Challenge (Habanero)

- Q1.** If $f(x) = x^2$ and $g(x) = 2x + 1$, what is $f(g(x))$?
(A) $4x^2 + 4x + 1$
(B) $4x^2 + 1$
(C) $2x^2 + 2x + 1$
(D) $x^2 + 2x + 1$
(E) $4x^2 + 2$
- Q2.** A hiker's fuel consumption is given by $f(x) = x/10$ gallons per mile, and her distance traveled is $g(x) = 2x + 5$ miles. What is $f(g(x))$?
(A) $(2x + 5)/10$
(B) $2x/10 + 5$
(C) $x/5 + 1$
(D) $2x + 5/10$
(E) $x/2 + 5/2$
- Q3.** Given $f(x) = x - 2$ and $g(x) = 3x + 1$, what is the domain of $f(g(x))$?
(A) All real numbers
(B) All real numbers except $x = -1$
(C) All real numbers except $x = 0$
(D) $x \geq -1$
(E) $x \leq 1$
- Q4.** The population of deer in a forest is given by $f(x) = 2x + 10$, where x is the number of years. If $g(x) = x - 2$ represents the number of years since a natural disaster, what is $f(g(x))$?
(A) $2x - 4 + 10$
(B) $2x + 10 - 2$
(C) $2x + 6$
(D) $2x - 4$
(E) $2x + 4$
- Q5.** A park ranger travels $f(x) = x^2 + 2x$ miles in x hours. If $g(x) = 3x - 1$ represents the number of hours worked, what is $f(g(x))$?
(A) $(3x - 1)^2 + 2(3x - 1)$
(B) $x^2 + 2x$
(C) $9x^2 - 6x + 1 + 6x - 2$
(D) $9x^2 - 2$
(E) $3x^2 + 2x$
- Q6.** Given $f(x) = \frac{1}{x}$ and $g(x) = 2x - 1$, what is the domain of $f(g(x))$?
(A) All real numbers except $x = 0$
(B) All real numbers except $x = 1/2$
(C) $x > 0$
(D) $x \geq 1$
(E) $x \leq 1$
- Q7.** The cost of hiking in dollars is given by $f(x) = x + 5$, where x is the number of hours. If $g(x) = 2x + 1$ represents the number of hours spent hiking, what is $f(g(x))$?
(A) $2x + 1 + 5$
(B) $2x + 6$
(C) $x + 5$
(D) $x + 6$
(E) $2x - 1$
- Q8.** A researcher studies the population of a specific animal species, given by $f(x) = x^2 + 3x + 2$. If $g(x) = x - 1$ represents the number of years since the last study, what is $f(g(x))$?
(A) $(x - 1)^2 + 3(x - 1) + 2$
(B) $x^2 + 2x + 1$
(C) $x^2 + 3x - 1$
(D) $x^2 + x$
(E) $x^2 - x + 1$

Open Ended Questions (FRQ)

- Q9.** A function $f(x)$ represents the amount of fuel a car consumes, in gallons, when traveling x miles. A second function $g(x) = 2x + 1$ gives the distance traveled, in miles, when the driver has been driving for x hours. Find $f(g(x))$ if $f(x) = \frac{x}{10}$. Show all work and explain your reasoning.
- Q10.** A park's population of birds is modeled by the function $f(x) = 3x + 10$, where x is the number of years since the park was established. If $g(x) = x - 2$ represents the number of years since a new species was introduced, find and interpret $f(g(x))$. Provide a real-world explanation for your answer.

Chef's Tips

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- Tip 1: Ensure students understand the concept of composite functions before taking the test.
- Tip 2: Remind students to show their work and explain their reasoning for free-response questions.
- Tip 3: Encourage students to read each question carefully and choose the correct answer.